



ERROR CODE			DM 1 STATUS LAMP	ERROR DESCRIPTION	TCON ACTION	DRIVER ACTION	FAILURE CAUSE	TROUBLESHOOTING
DANA [HEX]	SAE J1939							
	SPN	FMI						
84.00	120	0	Amber warning Flashing	Analog Input function: Transmission retarder temperature : Data Valid but Above Normal Operational Range - Most Severe	The TCON is protecting the transmission and does not allow that the retarder temperature exceeds its limit. The retarder function will be gradually to fully disabled	Apply the service brakes, and allow the retarder / transmission oil to cool down. Once the temperature is below the critical temperature, resume normal operation.	The measured value of the retarder temperature is above the critical temperature.	Check the retarder temperature and the retarder temperature sensor. If operating in dirty environment clean out radiator fins to allow maximum air flow.
84.03	120	3	Amber warning Flashing	Analog Input function: Transmission retarder temperature : Voltage Above Normal, or Shorted To High Source	The retarder temperature value will be replaced by an appropriate default value. The TCON will disable the retarder functionality.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
85.00	3027	0	Amber warning Flashing	Input function: Transmission oil level detection: Data Valid but Above Normal Operational Range - Most Severe	The TCON is detected that the transmission oil level is too high.	Check service manual how to drain oil from the transmission.	Oil level in the transmission is too high	- Check the transmission oil level - Check the wiring between the TCON and the oil level sensor - Check installation and operation of the oil level sensor
85.01	3027	1	Amber warning Flashing	Input function: Transmission oil level detection: Data Valid but Below Normal Operational Range - Most Severe	The TCON is detected that the transmission oil level is too low.	Check service manual how to add oil to the transmission.	Oil level in the transmission is too low	- Check the transmission oil level - Check the wiring between the TCON and the oil level sensor - Check installation and operation of the oil level sensor
85.02	3027	2	Amber warning Flashing	Input function: Transmission oil level detection: Data Erratic, Intermittent, or Incorrect	The TCON has detected that the transmission oil level sensor are not in line.	Check oil level sensors.	Oil level sensors are in fault	- Check the wiring between the TCON and the oil level sensor - Check installation and operation of the oil level sensor
32.00	520208	0	Lamp Status Depending On Corresponding Functional Error	Analogue input 0 (resistive) (ANIA0): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to resistive analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input fiunction	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
32.01	520208	1	Lamp Status Depending On Corresponding Functional Error	Analogue input 0 (resistive) (ANIA0): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to resistive analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
32.03	520208	3	Lamp Status Depending On Corresponding Functional Error	Analogue input 0 (resistive) (ANIA0): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to resistive analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
32.04	520208	4	Lamp Status Depending On Corresponding Functional Error	Analogue input 0 (resistive) (ANIA0): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to resistive analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
33.00	520209	0	Lamp Status Depending On Corresponding Functional Error	analogue input 1 (resistive) (ANIA1): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input fiunction	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.

33.01	520209	1	Lamp Status Depending On Corresponding Functional Error	analogue input 1 (resistive) (ANIA1): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
33.03	520209	3	Lamp Status Depending On Corresponding Functional Error	analogue input 1 (resistive) (ANIA1): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
33.04	520209	4	Lamp Status Depending On Corresponding Functional Error	analogue input 1 (resistive) (ANIA1): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
33.05	520209	5	Lamp Status Depending On Corresponding Functional Error	analogue input 1 (resistive) (ANIA1): Current Below Normal, or Open Circuit	TCON will continue normal operation, but the value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. This may result in reduced operation of the vehicle.	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	An analogue input function that needs to be calibrated is activated on analogue input 0 and it has not been calibrated yet.	- Perform a calibration of the connected device. - Refer to the other reported errors to identify the affected analogue input function.
34.00	520210	0	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
34.01	520210	1	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
34.03	520210	3	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
34.04	520210	4	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
34.0F	520210	15	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the highest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
34.11	520210	17	Lamp Status Depending On Corresponding Functional Error	analogue input C0 (ANIC0): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.

35.00	520211	0	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
35.01	520211	1	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
35.03	520211	3	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
35.04	520211	4	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
35.0F	520211	15	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the highest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
35.11	520211	17	Lamp Status Depending On Corresponding Functional Error	analogue input C1 (ANIC1): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
36.00	520212	0	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
36.01	520212	1	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
36.03	520212	3	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
36.04	520212	4	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.

36.0F	520212	15	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the highest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
36.11	520212	17	Lamp Status Depending On Corresponding Functional Error	analogue input C2 (ANIC2): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
37.00	520213	0	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
37.01	520213	1	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
37.03	520213	3	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
37.04	520213	4	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
37.0F	520213	15	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the highest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
37.11	520213	17	Lamp Status Depending On Corresponding Functional Error	analogue input C3 (ANIC3): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
38.00	520214	0	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
38.01	520214	1	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input 0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.

38.03	520214	3	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
38.04	520214	4	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input 0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
38.0F	520214	15	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
38.11	520214	17	Lamp Status Depending On Corresponding Functional Error	analogue input C4 (ANIC4): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input 0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
3B.00	520217	0	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input A0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
3B.01	520217	1	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the measurement of the analogue input function connected to analogue input A0 will be affected. This may result in reduced operation of the vehicle, the TCON action depends on configured analog input function	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
3B.03	520217	3	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Voltage Above Normal, or Shorted To High Source	The value of the analogue input function connected to analogue input A0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
3B.04	520217	4	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Voltage Below Normal, or Shorted To Low Source	The value of the analogue input function connected to analogue input A0 will be replaced by an appropriate default value. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function
3B.0F	520217	15	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Data Valid but Above Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input A0 will be clipped to the highest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
3B.11	520217	17	Lamp Status Depending On Corresponding Functional Error	analogue input A0 (ANIA0): Data Valid but Below Normal Operational Range - Least Severe	The value of the analogue input function connected to analogue input A0 will be clipped to the lowest value allowed. The TCON action depends on configured analog input function	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
3C.00	520218	0	Lamp Status Depending On Corresponding Functional Error	Speed input 0: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% below the value of what is should be	- Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
3C.01	520218	1	Lamp Status Depending On Corresponding Functional Error	Speed input 0: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% above the value of what is should be	- Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
3C.02	520218	2	Stop Lamp Solid	Speed input 0: Data Erratic, Intermittent, or Incorrect	TCON will revert to limp home mode.	Contact maintenance for troubleshooting	Direction reporting error of the speed sensor.	- Check the speed sensor - Check the wiring of the speed sensor
3C.05	520218	5	Lamp Status Depending On Corresponding Functional Error	Speed input 0: Current Below Normal, or Open Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function

3C.06	520218	6	Lamp Status Depending On Corresponding Functional Error	Speed input 0: Current Above Normal or Grounded Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3D.00	520219	0	Lamp Status Depending On Corresponding Functional Error	Speed input 1: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% below the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3D.01	520219	1	Lamp Status Depending On Corresponding Functional Error	Speed input 1: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% above the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3D.02	520219	2	Stop Lamp Solid	Speed input 1: Data Erratic, Intermittent, or Incorrect	TCON will revert to limphome mode.	Contact maintenance for troubleshooting	Direction reporting error of the speed sensor.	- Check the speed sensor - Check the wiring of the speed sensor
3D.05	520219	5	Lamp Status Depending On Corresponding Functional Error	Speed input 1: Current Below Normal, or Open Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3D.06	520219	6	Lamp Status Depending On Corresponding Functional Error	Speed input 1: Current Above Normal or Grounded Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3E.00	520220	0	Lamp Status Depending On Corresponding Functional Error	Speed input 2: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% below the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3E.01	520220	1	Lamp Status Depending On Corresponding Functional Error	Speed input 2: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% above the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3E.02	520220	2	Stop Lamp Solid	Speed input 2: Data Erratic, Intermittent, or Incorrect	TCON will revert to limphome mode.	Contact maintenance for troubleshooting	Direction reporting error of the speed sensor.	- Check the speed sensor - Check the wiring of the speed sensor
3E.05	520220	5	Lamp Status Depending On Corresponding Functional Error	Speed input 2: Current Below Normal, or Open Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3E.06	520220	6	Lamp Status Depending On Corresponding Functional Error	Speed input 2: Current Above Normal or Grounded Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3F.00	520221	0	Lamp Status Depending On Corresponding Functional Error	Speed input 3: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% below the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3F.01	520221	1	Lamp Status Depending On Corresponding Functional Error	Speed input 3: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring.	Contact maintenance for troubleshooting	The transmission ratio measured by the controller is more than 5% above the value of what is should be.	Check the transmission, in order to understand if and which clutch is slipping. Check the transmission ratio settings of the controller.
3F.02	520221	2	Stop Lamp Solid	Speed input 3: Data Erratic, Intermittent, or Incorrect	TCON will revert to limphome mode.	Contact maintenance for troubleshooting	Direction reporting error of the speed sensor.	- Check the speed sensor - Check the wiring of the speed sensor
3F.05	520221	5	Lamp Status Depending On Corresponding Functional Error	Speed input 3: Current Below Normal, or Open Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
3F.06	520221	6	Lamp Status Depending On Corresponding Functional Error	Speed input 3: Current Above Normal or Grounded Circuit	The value of the assigned speed input function, will be replaced by an appropriate default value. The TCON action depends on the assigned speed function.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input function
46.03	520224	3	Lamp Status Depending On Corresponding Functional Error	Analog Output 0: Voltage Above Normal, or Shorted To High Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission shutdown mode. In all other cases the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
46.04	520224	4	Lamp Status Depending On Corresponding Functional Error	Analog Output 0: Voltage Below Normal, or Shorted To Low Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limphome mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
46.05	520224	5	Lamp Status Depending On Corresponding Functional Error	Analog Output 0: Current Below Normal, or Open Circuit	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limphome mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function

[illegible]

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9C.05	520248	5	Stop Lamp Solid	Output Function: Modulated Lockup: Current Below Normal, or Open Circuit	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9C.06	520248	6	Stop Lamp Solid	Output Function: Modulated Lockup: Current Above Normal or Grounded Circuit	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected an overcurrent.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9C.08	520248	8	Stop Lamp Solid	Output Function: Modulated Lockup: Abnormal Frequency or Pulse Width or Period	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9D.03	520249	3	Stop Lamp Solid	Output Function: Brake valve: Voltage Above Normal, or Shorted To High Source	TCON will revert to transmission limphome mode.			
9D.04	520249	4	Stop Lamp Solid	Output Function: Brake valve: Voltage Below Normal, or Shorted To Low Source	TCON will revert to transmission limphome mode.			
9D.05	520249	5	Stop Lamp Solid	Output Function: Brake valve: Current Below Normal, or Open Circuit	TCON will revert to transmission limphome mode.			
9D.06	520249	6	Stop Lamp Solid	Output Function: Brake valve: Current Above Normal or Grounded Circuit	TCON will revert to transmission shutdown mode.			
9D.08	520249	8	Stop Lamp Solid	Output Function: Brake valve: Abnormal Frequency or Pulse Width or Period	TCON will revert to transmission limphome mode.			
9E.03	520250	3	Stop Lamp Flashing	Output Function: Modulated retarder output: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9E.04	520250	4	Stop Lamp Solid	Output Function: Modulated retarder output: Voltage Below Normal, or Shorted To Low Source	TCON will revert to transmission limphome mode. The retarder will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9E.05	520250	5	Stop Lamp Solid	Output Function: Modulated retarder output: Current Below Normal, or Open Circuit	TCON will revert to transmission limphome mode. The retarder will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9E.06	520250	6	Stop Lamp Solid	Output Function: Modulated retarder output: Current Above Normal or Grounded Circuit	TCON will revert to transmission limphome mode. The retarder will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected an overcurrent.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
9E.08	520250	8	Stop Lamp Solid	Output Function: Modulated retarder output: Abnormal Frequency or Pulse Width or Period	TCON will revert to transmission limphome mode. The retarder will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
A0.03	520252	3	Amber Warning Solid	Output Function: Warning Light: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A0.04	520252	4	Amber Warning Solid	Output Function: Warning Light: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A0.05	520252	5	Amber Warning Solid	Output Function: Warning Light: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A0.09	520252	9	Amber Warning Solid	Output Function: Warning Light: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.

A1.03	520253	3	Amber Warning Solid	Output Function: Digital Temperature indication: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A1.04	520253	4	Amber Warning Solid	Output Function: Digital Temperature indication: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A1.05	520253	5	Amber Warning Solid	Output Function: Digital Temperature indication: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A1.09	520253	9	Amber Warning Solid	Output Function: Digital Temperature indication: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN .	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed .	Check if the CAN message containing the output function is configured.
A2.00	520254	0	Amber warning Flashing	Output Function: 4WD/2WD: Data Valid but Above Normal Operational Range - Most Severe	TCON will try to actuate the output.	Contact maintenance for troubleshooting	The mechanical actuator is stuck engaged or the feedback line is faulty.	- Check if digital output has its expected value and if the mechanical actuator has made the switch. - Check if the digital feedback is switched and still connected to the controller
A2.01	520254	1	Amber warning Flashing	Output Function: 4WD/2WD: Data Valid but Below Normal Operational Range - Most Severe	TCON will try to actuate the output.	Contact maintenance for troubleshooting	The mechanical actuator is stuck disengaged or the feedback line is faulty.	- Check if digital output has its expected value and if the mechanical actuator has made the switch. - Check if the digital feedback is switched and still connected to the controller
A2.03	520254	3	Amber warning Flashing	Output Function: 4WD/2WD: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A2.04	520254	4	Amber warning Flashing	Output Function: 4WD/2WD: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A2.05	520254	5	Amber warning Flashing	Output Function: 4WD/2WD: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A2.09	520254	9	Amber warning Flashing	Output Function: 4WD/2WD: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN .	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed .	Check if the CAN message containing the output function is configured.
A3.00	520255	0	Amber warning Flashing	Output Function: Low/High range: Data Valid but Above Normal Operational Range - Most Severe	TCON will try to actuate the output.	Contact maintenance for troubleshooting	The mechanical actuator is stuck engaged or the feedback line is faulty.	- Check if digital output has its expected value and if the mechanical actuator has made the switch. - Check if the digital feedback is switched and still connected to the controller
A3.01	520255	1	Amber warning Flashing	Output Function: Low/High range: Data Valid but Below Normal Operational Range - Most Severe	TCON will try to actuate the output.	Contact maintenance for troubleshooting	The mechanical actuator is stuck disengaged or the feedback line is faulty.	- Check if digital output has its expected value and if the mechanical actuator has made the switch. - Check if the digital feedback is switched and still connected to the controller
A3.03	520255	3	Amber warning Flashing	Output Function: Low/High range: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A3.04	520255	4	Amber warning Flashing	Output Function: Low/High range: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A3.05	520255	5	Amber warning Flashing	Output Function: Low/High range: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A3.09	520255	9	Amber warning Flashing	Output Function: Low/High range: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN .	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed .	Check if the CAN message containing the output function is configured.
A4.03	520256	3	Amber Warning Solid	Output Function: Speed dependant output: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output

A4.04	520256	4	Amber Warning Solid	Output Function: Speed dependant output: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A4.05	520256	5	Amber Warning Solid	Output Function: Speed dependant output: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A4.09	520256	9	Amber Warning Solid	Output Function: Speed dependant output: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
A5.03	520257	3	Amber warning Flashing	Output Function: Parking brake: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A5.04	520257	4	Amber warning Flashing	Output Function: Parking brake: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A5.05	520257	5	Amber warning Flashing	Output Function: Parking brake: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A5.09	520257	9	Amber warning Flashing	Output Function: Parking brake: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
A6.03	520258	3	Amber warning Flashing	Output Function: Parking brake light: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A6.04	520258	4	Amber warning Flashing	Output Function: Parking brake light: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A6.05	520258	5	Amber warning Flashing	Output Function: Parking brake light: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A6.09	520258	9	Amber warning Flashing	Output Function: Parking brake light: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
A7.03	520259	3	Stop Lamp Solid	Output Function: On/Off Lockup: Voltage Above Normal, or Shorted To High Source	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the digital output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
A7.04	520259	4	Stop Lamp Solid	Output Function: On/Off Lockup: Voltage Below Normal, or Shorted To Low Source	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the digital output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
A7.05	520259	5	Stop Lamp Solid	Output Function: On/Off Lockup: Current Below Normal, or Open Circuit	TCON will revert to transmission limphome mode. Lockup will be disabled for further driving .	In transmission limphome mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the digital output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
A7.09	520259	9	Amber warning Flashing	Output Function: On/Off Lockup: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
A8.03	520260	3	Amber warning Flashing	Output Function: Exhaust: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output

A8.04	520260	4	Amber warning Flashing	Output Function: Exhaust: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A8.05	520260	5	Amber warning Flashing	Output Function: Exhaust: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A8.09	520260	9	Amber warning Flashing	Output Function: Exhaust: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
A9.03	520261	3	Amber warning Flashing	Output Function: Retarder On/Off - Selector: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but the retarder function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A9.04	520261	4	Amber warning Flashing	Output Function: Retarder On/Off - Selector: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but the retarder function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A9.05	520261	5	Amber warning Flashing	Output Function: Retarder On/Off - Selector: Current Below Normal, or Open Circuit	TCON will continue normal operation, but the retarder function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
A9.09	520261	9	Amber warning Flashing	Output Function: Retarder On/Off: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
AA.03	520262	3	Amber warning Flashing	Output Function: Towing: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AA.04	520262	4	Amber warning Flashing	Output Function: Towing: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AA.05	520262	5	Amber warning Flashing	Output Function: Towing: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AA.09	520262	9	Amber warning Flashing	Output Function: Towing: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
AB.03	520263	3	Amber warning Flashing	Output Function: Overspeeding Light: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AB.04	520263	4	Amber warning Flashing	Output Function: Overspeeding Light: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AB.05	520263	5	Amber warning Flashing	Output Function: Overspeeding Light: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
AB.09	520263	9	Amber warning Flashing	Output Function: Overspeeding Light: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
AC.03	520264	3	Stop Lamp Solid	Output Function: Forward clutch selector: Voltage Above Normal, or Shorted To High Source	TCON will revert to transmission limp home mode. This clutch will be disabled for further driving.	In transmission limp home mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this output. - Check installation and operation of the connected actuator.
AC.04	520264	4	Stop Lamp Solid	Output Function: Forward clutch selector: Voltage Below Normal, or Shorted To Low Source	TCON will revert to transmission limp home mode. This clutch will be disabled for further driving.	In transmission limp home mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this output. - Check installation and operation of the connected actuator.

[illegible]

[illegible]

85.08	520273	8	Stop Lamp Solid	Output Function: Clutch High: Abnormal Frequency or Pulse Width or Period	TCON will revert to transmission limp mode. This clutch will be disabled for further driving .	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
88.03	520276	3	Amber warning Flashing	Output Function: Reverse Alert output: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
88.04	520276	4	Amber warning Flashing	Output Function: Reverse Alert output: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
88.05	520276	5	Amber warning Flashing	Output Function: Reverse Alert output: Current Below Normal, or Open Circuit	TCON will continue normal operation, but this output function will be disabled.	Contact maintenance for troubleshooting	The digital output assigned to this output function is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this digital output. - Check installation and operation of the connection. - Refer to the other reported errors to get more information about the digital output
88.09	520276	9	Amber warning Flashing	Output Function: Reverse Alert output: Abnormal Update Rate	TCON will continue normal operation, but this output function will not be broadcasted over CAN.	Contact maintenance for troubleshooting	CAN message containing value for the output function cannot be broadcasted or constructed.	Check if the CAN message containing the output function is configured.
89.00	520277	0	Amber warning Flashing	Torque Limiter: Data Valid but Above Normal Operational Range - Most Severe	TCON will force the torque limiter clutch open. This will result in inoperability of the vehicle. TCON will only resume normal operation after the shifter is cycled through neutral.	Contact maintenance for troubleshooting	The slip in the torque limiter has exceeded the maximum allowed time.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
89.03	520277	3	Stop Lamp Flashing	Torque Limiter: Clutch High: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
89.04	520277	4	Stop Lamp Flashing	Torque Limiter: Clutch High: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
89.05	520277	5	Stop Lamp Flashing	Torque Limiter: Clutch High: Current Below Normal, or Open Circuit	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected that the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
89.06	520277	6	Stop Lamp Flashing	Torque Limiter: Clutch High: Current Above Normal or Grounded Circuit	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	Dynamic checking by the TCON of the analog output detected an overcurrent.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
89.08	520277	8	Stop Lamp Flashing	Torque Limiter: Clutch High: Abnormal Frequency or Pulse Width or Period	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. Contact maintenance for troubleshooting .	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
BA.00	520278	0	Amber warning Flashing	Engine speed / torque exceeds allowed max speed / torque: Data Valid but Above Normal Operational Range - Most Severe.	TCON will force the transmission in neutral as protection mechanism, as too much torque going through the transmission can cause damage. TCON will only resume normal operation after the shifter is cycled through neutral.	Reduce engine speed or power. Recycle shifter through neutral to reset the protection mechanism.	Engine speed or torque is above commanded engine speed or torque limit for too long . Transmission protection forced neutral	- Check if engine control is following engine command provided by TCON - check CAN communication
BA.0F	520278	15	Protection Lamp	Engine speed / torque exceeds allowed max speed / torque: Data Valid but Above Normal Operational Range - Least Severe.	TCON will continue normal operation. If the overtorque or overspeed condition persists, TCON will escalate this warning into FMI 0.	Reduce engine speed or power.	Engine speed or torque is above commanded engine speed or torque limit for limited duration.	- Check if engine control is following engine command provided by TCON - check CAN communication
28.02	520286	2	Stop Lamp Flashing	direction selection: shift lever: Data Erratic, Intermittent, or Incorrect	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the direction selection contains an invalid pattern.	Check the wiring between the TCON and the shifter concerning the direction signals.
28.09	520286	9	Stop Lamp Flashing	direction selection: shift lever: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	CAN message containing value for the shifter function not received within the expected timeframe or not present on the CAN bus .	- Check if CAN message containing the shifter function value is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.

28.13	520286	19	Stop Lamp Flashing	direction selection: shift lever: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the shiftlever function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the shiftlever function for requesting a valid value.
29.02	520287	2	Amber warning Flashing	gear selection: shift lever: Data Erratic, Intermittent, or Incorrect	TCON will continue normal operation, but the value of the gear shift lever will be replaced by the closest corresponding value. This may result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the gear selection contains an invalid pattern.	Check the wiring between the TCON and the shiftlever concerning the gear signals.
29.09	520287	9	Amber warning Flashing	gear selection: shift lever: Abnormal Update Rate	TCON will continue normal operation, but the value of the gear shift lever will be replaced by the closest corresponding value. This may result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	CAN message containing value for the shiftlever function not received within the expected timeframe or not present on the CAN bus.	- Check if CAN message containing the shiftlever function value is sent correctly on the CAN bus. - Check if the repetition rate of the CAN message is fast enough.
29.13	520287	19	Amber warning Flashing	gear selection: shift lever: Received Network Data in Error	TCON will continue normal operation, but the value of the gear shift lever will be replaced by the closest corresponding value. This may result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the shiftlever function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the shiftlever function for requesting a valid value.
2D.09	520288	9	Amber warning Flashing	Digital Input Function: Shift lever Selection Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
2D.13	520288	19	Amber warning Flashing	Digital Input Function: Shift lever Selection Received Network Data in Error	TCON will continue normal operation, but the value of the input function will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
4D.03	520294	3	Lamp Status Depending On Corresponding Functional Error	Analog Output 6: Voltage Above Normal, or Shorted To High Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission shutdown mode. In all other cases the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4D.04	520294	4	Lamp Status Depending On Corresponding Functional Error	Analog Output 6: Voltage Below Normal, or Shorted To Low Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4D.05	520294	5	Lamp Status Depending On Corresponding Functional Error	Analog Output 6: Current Below Normal, or Open Circuit	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4D.06	520294	6	Lamp Status Depending On Corresponding Functional Error	Analog Output 6: Current Above Normal or Grounded Circuit	If the output has a critical transmission control function assigned to it, TCON will revert to transmission shutdown mode. In all other cases the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected an overcurrent.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4D.08	520294	8	Lamp Status Depending On Corresponding Functional Error	Analog Output 6: Abnormal Frequency or Pulse Width or Period	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4E.03	520295	3	Lamp Status Depending On Corresponding Functional Error	Analog Output 7: Voltage Above Normal, or Shorted To High Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission shutdown mode. In all other cases the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected that the output is shorted to battery plus (high source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4E.04	520295	4	Lamp Status Depending On Corresponding Functional Error	Analog Output 7: Voltage Below Normal, or Shorted To Low Source	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the output detected that the output is shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.
4E.05	520295	5	Lamp Status Depending On Corresponding Functional Error	Analog Output 7: Current Below Normal, or Open Circuit	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation.	In transmission limp mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. Contact maintenance for troubleshooting	Dynamic checking by the TCON of the analog output detected the output is in open circuit (loose connection).	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function.

4E.06	520295	6	Lamp Status Depending On Corresponding Functional Error	Analog Output 7: Current Above Normal or Grounded Circuit	If the output has a critical transmission control function assigned to it, TCON will revert to transmission shutdown mode. In all other cases the assigned function control will be disabled, possibly resulting in reduced vehicle operation. <u>Contact maintenance for troubleshooting</u>	In transmission shutdown mode, the driver can no longer operate the vehicle. Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. <u>Contact maintenance for troubleshooting</u>	Dynamic checking by the TCON of the analog output detected an overcurrent.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
4E.08	520295	8	Lamp Status Depending On Corresponding Functional Error	Analog Output 7: Abnormal Frequency or Pulse Width or Period	If the output has a critical transmission control function assigned to it, TCON will revert to transmission limp home mode. In any case the assigned function control will be disabled, possibly resulting in reduced vehicle operation. <u>Contact maintenance for troubleshooting</u>	In transmission limp home mode, the driver can still operate the vehicle (reduced vehicle operation). Return to a service center for troubleshooting as soon as possible. In all other cases disabling the assigned function control will result in reduced vehicle operation. <u>Contact maintenance for troubleshooting</u>	The TCON detected that the output current is out of range. This occurs when the load has an incorrect impedance.	- Check the wiring between the TCON and the device connected to this analog output. - Check installation and operation of the connected actuator. - refer to the other reported errors to get more information about the configured output function
15.00	520297	0	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Data Valid but Above Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
15.01	520297	1	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Data Valid but Below Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
15.03	520297	3	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Voltage Above Normal, or Shorted To High Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
15.04	520297	4	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Voltage Below Normal, or Shorted To Low Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
15.0F	520297	15	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
15.11	520297	17	Lamp Status Depending On Corresponding Functional Error	Analog input B0 (ANIB0): Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
16.00	520298	0	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Data Valid but Above Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
16.01	520298	1	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Data Valid but Below Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
16.03	520298	3	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Voltage Above Normal, or Shorted To High Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
16.04	520298	4	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Voltage Below Normal, or Shorted To Low Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
16.0F	520298	15	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
16.11	520298	17	Lamp Status Depending On Corresponding Functional Error	Analog input B1 (ANIB1): Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
17.00	520299	0	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Data Valid but Above Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
17.01	520299	1	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Data Valid but Below Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
17.03	520299	3	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Voltage Above Normal, or Shorted To High Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
17.04	520299	4	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Voltage Below Normal, or Shorted To Low Source	Depending on the configured input, TCON will take the necessary action. Refer to the <u>functional error of the configured input</u> .	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.

17.0F	520299	15	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
17.11	520299	17	Lamp Status Depending On Corresponding Functional Error	Analog input B2 (ANIB2): Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
18.00	520300	0	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Data Valid but Above Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
18.01	520300	1	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Data Valid but Below Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
18.03	520300	3	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Voltage Above Normal, or Shorted To High Source	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
18.04	520300	4	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Voltage Below Normal, or Shorted To Low Source	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
18.0F	520300	15	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
18.11	520300	17	Lamp Status Depending On Corresponding Functional Error	Analog input B3 (ANIB3): Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
19.00	520301	0	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Data Valid but Above Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
19.01	520301	1	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Data Valid but Below Normal Operational Range - Most Severe	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
19.03	520301	3	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Voltage Above Normal, or Shorted To High Source	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
19.04	520301	4	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Voltage Below Normal, or Shorted To Low Source	Depending on the configured input, TCON will take the necessary action. Refer to the functional error of the configured input.	Contact maintenance for troubleshooting	Refer to the functional error of the configured input.	Refer to the functional error of the configured input.
19.0F	520301	15	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
19.11	520301	17	Lamp Status Depending On Corresponding Functional Error	Analog input B4 (ANIB4): Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
9.00	520303	0	Stop Lamp Flashing	Input Function: Lube Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high: - Bad wire - Hydraulic failure	- Check lube pressure at engine idle. - Log pressure (CAN trace or other measurement equipment) when problem occurs.
9.01	520303	1	Stop Lamp Flashing	Input Function: Lube Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low: - Bad wire - Hydraulic failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. - Log pressure (CAN trace or other measurement equipment) when problem occurs.

9.03	520303	3	Stop Lamp Flashing	Input Function: Lube Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
9.04	520303	4	Stop Lamp Flashing	Input Function: Lube Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
9.09	520303	9	Stop Lamp Flashing	Input Function: Lube Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
9.13	520303	19	Stop Lamp Flashing	Input Function: Lube Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
0A.00	520304	0	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	Default action: TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted. Alternative action: The TCON will revert to transmission limp/home mode.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
0A.01	520304	1	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
0A.03	520304	3	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0A.04	520304	4	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0A.09	520304	9	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
0A.13	520304	19	Stop Lamp Flashing	Input Function: Forward Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
0B.00	520305	0	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
0B.01	520305	1	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
0B.03	520305	3	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0B.04	520305	4	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0B.09	520305	9	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.

OB.13	520305	19	Stop Lamp Flashing	Input Function: Reverse Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
OC.00	520306	0	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
OC.01	520306	1	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
OC.03	520306	3	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
OC.04	520306	4	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
OC.09	520306	9	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
OC.13	520306	19	Stop Lamp Flashing	Input Function: Forward High Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
OD.00	520307	0	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
OD.01	520307	1	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
OD.03	520307	3	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
OD.04	520307	4	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
OD.09	520307	9	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
OD.13	520307	19	Stop Lamp Flashing	Input Function: 1st Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
OE.00	520308	0	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
OE.01	520308	1	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.

0E.03	520308	3	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0E.04	520308	4	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0E.09	520308	9	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
0E.13	520308	19	Stop Lamp Flashing	Input Function: 2nd Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
0F.00	520309	0	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
0F.01	520309	1	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
0F.03	520309	3	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0F.04	520309	4	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
0F.09	520309	9	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
0F.13	520309	19	Stop Lamp Flashing	Input Function: 3rd Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
10.00	520310	0	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
10.01	520310	1	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
10.03	520310	3	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
10.04	520310	4	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
10.09	520310	9	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.

10.13	520310	19	Stop Lamp Flashing	Input Function: 4th Clutch Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
11.00	520311	0	Stop Lamp Flashing	Input Function: Retarder Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	By using the retarder diagnostic mode: - Check pressure when retarder is NOT selected (+0 bar required). - Check pressure when retarder is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
11.01	520311	1	Stop Lamp Flashing	Input Function: Retarder Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	By using the retarder diagnostic mode: - Check pressure when retarder is NOT selected (+0 bar required). - Check pressure when retarder is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
11.03	520311	3	Stop Lamp Flashing	Input Function: Retarder Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus.	check wiring and voltage state of the pins of the retarder pressure sensor Log pressure (CAN trace or other measurement equipment) when problem occurs.
11.04	520311	4	Stop Lamp Flashing	Input Function: Retarder Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	check wiring and voltage state of the pins of the retarder pressure sensor Log pressure (CAN trace or other measurement equipment) when problem occurs.
11.09	520311	9	Stop Lamp Flashing	Input Function: Retarder Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
11.13	520311	19	Stop Lamp Flashing	Input Function: Retarder Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
45.00	520312	0	Amber warning Flashing	Lockup Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but lockup will be unavailable. TCON will only try to reactivate lockup when TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+0 bar required). - Check clutch pressure when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
45.01	520312	1	Amber warning Flashing	Lockup Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but lockup will be unavailable. TCON will only try to reactivate lockup when TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs.
45.03	520312	3	Amber warning Flashing	Lockup Pressure: Voltage Above Normal, or Shorted To High Source	TCON will continue normal operation but lockup will be unavailable. TCON will only try to reactivate lockup when TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
45.04	520312	4	Amber warning Flashing	Lockup Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will continue normal operation but lockup will be unavailable. TCON will only try to reactivate lockup when TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
45.07	520312	7	Amber warning Flashing	Lockup engaged but turbine and impeller have speed difference	TCON will continue to use the lockup logic. However, this might result in reduced operability of the vehicle.	Contact maintenance for troubleshooting	- The lockup valve has no sufficient pressure. - Lockup clutch failure	- Check pressure lockup valve - Check lockup actuator
45.09	520312	9	Amber warning Flashing	Lockup Pressure: Abnormal Update Rate	TCON will continue normal operation but lockup will be unavailable.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
45.13	520312	19	Amber warning Flashing	Lockup Pressure: Received Network Data in Error	TCON will continue normal operation but lockup will be unavailable.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
2E.0F	520313	15	Amber Warning Solid	Inching torque: Data Valid but Above Normal Operational Range - Least Severe	TCON will open or close the inching clutch depending of the GDE parameter selection.	Release the inching pedal and cycle the shifter lever via neutral back in the selected direction in order to get drive again.	The transmission has exceeded the maximum torque during inching.	- probably no servicing action needed, just temporary reduction of vehicle usage intensity to allow transmission to cool down. - if problem persists, Check functioning of transmission control.
75.01	520314	1	Stop Lamp Flashing	Speed Sensor Input Function: Auxiliary motor speed: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	The auxiliary motor speed is too low to guarantee safe operation. Transmission can become damaged due to insufficient auxiliary motor speed.	- Check the wiring between the TCON and CAN interface. - Check the source controller that is sending this signal. - refer to the other reported errors to identify the affected speed input.

75.09	520314	9	Stop Lamp Flashing	Speed Sensor Input Function: Auxiliary motor speed: Abnormal Update Rate	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	The auxiliary motor speed is missing and therefore the TCON can't guarantee safe operation. Transmission can become damaged due to insufficient auxiliary motor speed.	- Check if CAN message containing the intermediate speed value is sent correctly on the CAN bus . - Check if the repetition rate of that CAN message is fast enough. - Check CAN connection to the controller.
75.13	520314	19	Stop Lamp Flashing	Speed Sensor Input Function: Auxiliary motor speed: Received Network Data in Error	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	The auxiliary motor speed is missing and therefore the TCON can't guarantee safe operation. Transmission can become damaged due to insufficient auxiliary motor speed.	- Check source of broadcast. - Check CAN connection to the controller.
2F.00	520315	0	Amber warning Flashing	Transmission limit: Data Valid but Above Normal Operational Range - Most Severe	TCON detected overspeeding conditions . Depending on the settings, the controller might upshift to a higher gear and eventually go to a neutral zero state. The error code remains active until the corrective action is reset.	Release the throttle pedal, and apply brakes.	Transmission input or output speed is too high.	-slow down vehicle by applying brakes or releasing throttle pedal
41.00	520316	0	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Data Valid but Above Normal Operational Range - Most Severe	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	Engine speed is above commanded engine speed limit for 2 seconds .	Check can connection to the VSM. Run trace when problem occurs. Check for engine speed response on limit request by controller.
41.01	520316	1	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Data Valid but Below Normal Operational Range - Most Severe	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	Turbine speed was detected but no engine speed was measured.	- Check the transmission and speed sensors. - Check the speed sensor position. - Check the can signal containing engine speed (if configured via can)
41.05	520316	5	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Current Below Normal, or Open Circuit	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the transmission and speed sensors. - Check the speed sensor position. - Check the can signal containing engine speed (if configured via can)
41.06	520316	6	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Current Above Normal or Grounded Circuit	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the device connected to this speed input. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input.
41.09	520316	9	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Abnormal Update Rate	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	No engine speed information available, as the EEC1 or CVC2TC message is missing on the CAN bus .	- Check if CAN message containing the engine speed value is sent correctly on the CAN bus . - Check if the repetition rate of that CAN message is fast enough. - Check CAN connection to the controller.
41.13	520316	19	Stop Lamp Solid	Speed Sensor Input Function: Engine speed: Received Network Data in Error	The value of the engine speed sensor, will be replaced by an appropriate calculated value. The TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	The contents of the engine speed field in the EEC1 or CVC2TC message is set to error or not available.	- Check engine controller. - Check CAN connection to the controller.
42.00	520317	0	Amber Warning Solid	Speed Sensor Input Function: Turbine speed: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The turbine rpm is 100 rpm above what it should be. This can be determined using drum and output speed combined with the gear ratios. This check can only be done when the turbine is connected to the output.	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
42.01	520317	1	Amber Warning Solid	Speed Sensor Input Function: Turbine speed: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The turbine rpm is 100 rpm below what it should be. This can be determined using drum and output speed combined with the gear ratios. This check can only be done when the turbine is connected to the output.	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
42.02	520317	2	Stop Lamp Solid	Speed Sensor Input Function: Turbine speed: Data Erratic, Intermittent, or Incorrect	TCON will revert to limp home mode.	Contact maintenance for troubleshooting	Direction reporting error of the speed sensor.	- Check the speed sensor - Check the wiring of the speed sensor
42.05	520317	5	Amber Warning Solid	Speed Sensor Input Function: Turbine speed: Current Below Normal, or Open Circuit	The value of the turbine speed sensor, will be replaced by an appropriate calculated value. The TCON will disable inching	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the turbine speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input.
42.06	520317	6	Amber Warning Solid	Speed Sensor Input Function: Turbine speed: Current Above Normal or Grounded Circuit	The value of the turbine speed sensor, will be replaced by an appropriate calculated value. The TCON will disable inching	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the turbine speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input.
43.00	520318	0	Amber Warning Solid	Speed Sensor Input Function: intermediate speed: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The intermediate rpm is 100 rpm above what it should be. This can be determined using output and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output.	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
43.01	520318	1	Amber Warning Solid	Speed Sensor Input Function: intermediate speed: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The intermediate rpm is 100 rpm below what it should be. This can be determined using output and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output.	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.

43.05	520318	5	Amber Warning Solid	Speed Sensor Input Function: intermediate speed: Current Below Normal, or Open Circuit	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the turbine speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input
43.06	520318	6	Amber Warning Solid	Speed Sensor Input Function: intermediate speed: Current Above Normal or Grounded Circuit	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the intermediate speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input
44.00	520319	0	Amber Warning Solid	Speed Sensor Input Function: Output speed: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The output rpm is 100 rpm above what it should be. This can be determined using drum and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
44.01	520319	1	Amber Warning Solid	Speed Sensor Input Function: Output speed: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The output rpm is 100 rpm below what it should be. This can be determined using drum and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
44.05	520319	5	Amber Warning Solid	Speed Sensor Input Function: Output speed: Current Below Normal, or Open Circuit	The value of the output speed sensor, will be replaced by an appropriate calculated value. The TCON will disable inching	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the output speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input
44.06	520319	6	Amber Warning Solid	Speed Sensor Input Function: Output speed: Current Above Normal or Grounded Circuit	The value of the output speed sensor, will be replaced by an appropriate calculated value. The TCON will disable inching	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the output speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input
2A.00	520320	0	Stop Lamp Flashing	transmission ratio: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A.01	520320	1	Stop Lamp Flashing	transmission ratio: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 14	520320	20	Stop Lamp Solid	transmission ratio: Forward Clutch Slip - Most Severe	Forward Clutch Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 15	520320	21	Stop Lamp Solid	transmission ratio: Reverse Clutch Slip - Most Severe	Reverse Clutch Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 16	520320	22	Stop Lamp Solid	transmission ratio: Forward High Clutch Slip - Most Severe	Forward High Clutch Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 17	520320	23	Stop Lamp Solid	transmission ratio: Stage 1 clutch D Slip - Most Severe	Stage 1 clutch D Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 18	520320	24	Stop Lamp Solid	transmission ratio: 1st Clutch Slip - Most Severe	1st Clutch Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 19	520320	25	Stop Lamp Solid	transmission ratio: 2nd Clutch Slip - Most Severe	2nd Clutch Slip - Most Severe TCON will revert to limp mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors

2A. 1A	520320	26	Stop Lamp Solid	transmission ratio: 3rd Clutch Slip - Most Severe	3rd Clutch Slip - Most Severe TCON will revert to limp home mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 1B	520320	27	Stop Lamp Solid	transmission ratio: 4th Clutch Slip - Most Severe	4th Clutch Slip - Most Severe TCON will revert to limp home mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 1C	520320	28	Stop Lamp Solid	transmission ratio: High Clutch Slip - Most Severe	High Clutch Slip - Most Severe TCON will revert to limp home mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is above expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check oil level - check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2A. 1D	520320	29	Stop Lamp Solid	transmission ratio: Low Clutch Slip - Most Severe	Low Clutch Slip - Most Severe TCON will revert to limp home mode. Transmission will go into neutral until standstill. After that, the lowest safe gear can be engaged.	Contact maintenance for troubleshooting	Transmission Gear Ratio calculated from measured speed signals is below expected value for current gear. Uncontrolled slip in one or more clutches is possible.	- check transmission system pressure - check clutch pressure of all clutches with different gears engaged - check stall speed with different gears engaged - check speed signals of all speed sensors
2C.00	520321	0	Amber Warning Solid	Analog Input Function: Throttle Pedal: Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the throttle pedal will be clipped to the highest value allowed.	Contact maintenance for troubleshooting, or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input
2C.01	520321	1	Amber Warning Solid	Analog Input Function: Throttle Pedal: Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the throttle pedal will be clipped to the lowest value allowed.	Contact maintenance for troubleshooting, or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input
2C.03	520321	3	Amber warning Flashing	Analog Input Function: Throttle Pedal: Voltage Above Normal, or Shorted To High Source	The value of the throttle pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no shifting or inching will be allowed.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input
2C.04	520321	4	Amber warning Flashing	Analog Input Function: Throttle Pedal: Voltage Below Normal, or Shorted To Low Source	The value of the throttle pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no shifting or inching will be allowed.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input
2C.09	520321	9	Amber warning Flashing	Analog Input Function: Throttle Pedal: Abnormal Update Rate	The TCON will flag the fault and the TSC1 message will send the idle position to the engine controller. The value of the throttle pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no shifting or inching will be allowed.	Contact maintenance for troubleshooting	No throttle pedal information available, as the EEC2 message is missing on the CAN bus.	- Check if CAN message containing the throttle pedal value is sent correctly on the CAN bus. - Check if the repetition rate of that CAN message is fast enough. - Check CAN connection to the controller.
2C.0F	520321	15	Amber Warning Solid	Analog Input Function: Throttle Pedal: Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
2C. 11	520321	17	Amber Warning Solid	Analog Input Function: Throttle Pedal: Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
2C. 13	520321	19	Amber warning Flashing	Analog Input Function: Throttle Pedal: Received Network Data in Error	The TCON will flag the fault and the TSC1 message will send the idle position to the engine controller. The value of the throttle pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no shifting or inching will be allowed.	Contact maintenance for troubleshooting	The contents of the throttle pedal field in the EEC2 message is set to error or not available.	- Check throttle pedal broadcast source. - Check CAN connection to the controller.
70.00	520322	0	Amber Warning Solid	Analog Input Function: Hydro Lever: Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the hydro lever will be clipped to the highest value allowed.	Contact maintenance for troubleshooting, or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input

70.01	520322	1	Amber Warning Solid	Analog Input Function: Hydro Lever: Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the hydro lever will be clipped to the lowest value allowed.	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the hydro lever. - Check installation and operation of the hydro lever. - refer to the other reported errors to identify the affected analogue input
70.03	520322	3	Amber warning Flashing	Analog Input Function: Hydro Lever: Voltage Above Normal, or Shorted To High Source	The TCON will flag the fault and the value of the hydro lever will be replaced by an appropriate default value. This will result in reduced vehicle operation, no engine control for hydro lever will be available.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the hydro lever. - Check installation and operation of the hydro lever. - refer to the other reported errors to identify the affected analogue input
70.04	520322	4	Amber warning Flashing	Analog Input Function: Hydro Lever: Voltage Below Normal, or Shorted To Low Source	The TCON will flag the fault and the value of the hydro lever will be replaced by an appropriate default value. This will result in reduced vehicle operation, no engine control for hydro lever will be available.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the hydro lever. - Check installation and operation of the hydro lever. - refer to the other reported errors to identify the affected analogue input
70.09	520322	9	Amber warning Flashing	Analog Input Function: Hydro Lever: Abnormal Update Rate	The TCON will flag the fault and the value of the hydro lever will be replaced by an appropriate default value. This will result in reduced vehicle operation, no engine control for hydro lever will be available	Contact maintenance for troubleshooting	No hydro lever information available, as the required CVC2TC message is missing on the CAN bus .	- Check if CAN message containing the hydro lever value is sent correctly on the CAN bus . - Check if the repetition rate of that CAN message is fast enough.
70.0F	520322	15	Amber Warning Solid	Analog Input Function: Hydro Lever: Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check CAN connection to the controller - Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
70.11	520322	17	Amber Warning Solid	Analog Input Function: Hydro Lever: Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input
70.13	520322	19	Amber warning Flashing	Analog Input Function: Hydro Lever: Received Network Data in Error	The TCON will flag the fault and the value of the hydro lever will be replaced by an appropriate default value. This will result in reduced vehicle operation, no engine control for hydro lever will be available	Contact maintenance for troubleshooting	The contents of the hydro lever field in the CVC2TC message is set to error or not available.	- Check hydro lever broadcast source. - Check CAN connection to the controller.
71.00	520323	0	Amber Warning Solid	Analog Input Function: Brake Pedal: Data Valid but Above Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the brake pedal will be clipped to the highest value allowed.	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is above the maximum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the throttle pedal. - Check installation and operation of the throttle pedal. - refer to the other reported errors to identify the affected analogue input
71.01	520323	1	Amber Warning Solid	Analog Input Function: Brake Pedal: Data Valid but Below Normal Operational Range - Most Severe	TCON will continue normal operation but the value for the brake pedal will be clipped to the lowest value allowed.	Contact maintenance for troubleshooting or Perform a calibration of the connected device.	The measured value of the analog input is below the minimum allowed range.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the brake pedal. - Check installation and operation of the brake pedal. - refer to the other reported errors to identify the affected analogue input
71.03	520323	3	Amber warning Flashing	Analog Input Function: Brake Pedal: Voltage Above Normal, or Shorted To High Source	The value of the brake pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no inching will be allowed.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the brake pedal. - Check installation and operation of the brake pedal. - refer to the other reported errors to identify the affected analogue input
71.04	520323	4	Amber warning Flashing	Analog Input Function: Brake Pedal: Voltage Below Normal, or Shorted To Low Source	The value of the brake pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no inching will be allowed.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Recalibrate the input, if the problem persists, check the wiring between the TCON and the brake pedal. - Check installation and operation of the brake pedal. - refer to the other reported errors to identify the affected analogue input
71.09	520323	9	Amber warning Flashing	Analog Input Function: Brake Pedal: Abnormal Update Rate	The TCON will flag the fault and the value of the brake pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no inching will be allowed.	Contact maintenance for troubleshooting	No brake pedal information available, as the required CVC2TC message is missing on the CAN bus .	- Check if CAN message containing the brake pedal value is sent correctly on the CAN bus . - Check if the repetition rate of that CAN message is fast enough.
71.0F	520323	15	Amber Warning Solid	Analog Input Function: Brake Pedal: Data Valid but Above Normal Operational Range - Least Severe	TCON will show the error code, and will use the highest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check CAN connection to the controller - Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input

71.11	520323	17	Amber Warning Solid	Analog Input Function: Brake Pedal: Data Valid but Below Normal Operational Range - Least Severe	TCON will show the error code, and will use the lowest clipped value in its settings for this analog input.	Contact maintenance for troubleshooting	The input is out of calibration.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the connected device. - refer to the functional error of the attached device. - Perform a calibration of the device attached to the input.
71.13	520323	19	Amber warning Flashing	Analog Input Function: Brake Pedal: Received Network Data in Error	The TCON will flag the fault and the value of the brake pedal will be replaced by an appropriate default value. This will result in reduced vehicle operation, possibly no inching will be allowed.	Contact maintenance for troubleshooting	The contents of the brake pedal field in the CVC2TC message is set to error or not available.	- Check brake pedal broadcast source. - Check CAN connection to the controller.
72.00	520324	0	Amber warning Flashing	Analog Input Function: Transmission Sump Temperature: Data Valid but Above Normal Operational Range - Most Severe	The TCON will flag the fault, and take the configured action (either no action, force neutral or reduce engine revs)	Contact maintenance for troubleshooting	Transmission sump temperature too hot.	- Check the wiring between the TCON and the sump temperature sensor. - Check installation and operation of the sump temperature sensor. - refer to the other reported errors to identify the configured analogue input.
72.03	520324	3	Amber warning Flashing	Analog Input Function: Transmission Sump Temperature: Voltage Above Normal, or Shorted To High Source	The TCON will flag the fault and will use a default value in its settings (which results in bad temperature compensation)	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the operation of the cooler - Check the wiring between the TCON and the sump temperature sensor. - Check installation and operation of the sump temperature sensor. - refer to the other reported errors to identify the configured analogue input.
72.04	520324	4	Amber warning Flashing	Analog Input Function: Transmission Sump Temperature: Voltage Below Normal, or Shorted To Low Source	The TCON will flag the fault and will use a default value in its settings (which results in bad temperature compensation)	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the sump temperature sensor. - Check installation and operation of the sump temperature sensor. - refer to the other reported errors to identify the configured analogue input.
72.0F	520324	15	Protection Lamp	Analog Input function: Transmission Sump out temperature : Data Valid but Above Normal Operational Range - Least Severe	TCON will flag the fault in order to indicate the warning level to the driver.	Take notice of the fault, and try to bring the sump temperature to a lower level.	The measured value of the sump temperature is above the warning level temperature.	Check the sump temperature and the converter out temperature sensor.
28.00	520325	0	Amber warning Flashing	Analog Input function: Transmission convertor out temperature : Data Valid but Above Normal Operational Range - Most Severe	The TCON is protecting the transmission and does not allow that the converter out temperature exceeds its limit. The engine will be limited to 1500 rpm.	Stop the vehicle, select neutral and apply full throttle (limited to 1500 rpm) in order to cool the transmission oil until the fault disappears. Once the converter out temperature is below the critical temperature, resume normal operation.	The measured value of the convertor out temperature is above the critical temperature.	Check the converter out temperature and the converter out temperature sensor. If operating in dirty environment clean out radiator fins to allow maximum air flow.
28.03	520325	3	Amber warning Flashing	Analog Input function: Transmission convertor out temperature : Voltage Above Normal, or Shorted To High Source	The convertor out temperature value will be replaced by an appropriate default value. The TCON will continue normal operation.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
28.04	520325	4	Amber warning Flashing	Analog Input function: Transmission convertor out temperature : Voltage Below Normal, or Shorted To Low Source	The convertor out temperature value will be replaced by an appropriate default value. The TCON will continue normal operation.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the device connected to this analog input. - Check installation and operation of the device connected. - refer to the other reported errors to identify the affected analogue input function.
28.0F	520325	15	Protection Lamp	Analog Input function: Transmission convertor out temperature : Data Valid but Above Normal Operational Range - Least Severe	TCON will flag the fault in order to indicate the warning level to the driver.	Take notice of the fault, and try to bring the converter temperature to a lower level.	The measured value of the convertor out temperature is above the warning level temperature.	Check the converter out temperature and the converter out temperature sensor.
73.00	520326	0	Amber warning Flashing	Input Function: Filter Clogging: Data Valid but Above Normal Operational Range - Most Severe	TCON will disable inching . Other actions depends on parameter setting: - either TCON will limit the engine speed to a lower value until the filter clogging input becomes inactive. - either TCON will take no further action	Replace filter or Contact maintenance for troubleshooting	- Filter is contaminated - The measured value of the analog input is above the maximum allowed range. - The digital inputs (NO and NC) are both in the on state	- replace the filter - check the wiring between the TCON and the filter indicator. - Check installation and operation of the filter indicator. - refer to the other reported errors to identify the affected analogue input
73.03	520326	3	Amber warning Flashing	Input Function: Filter Clogging: Voltage Above Normal, or Shorted To High Source	TCON will disable inching . Other actions depends on parameter setting: - either TCON will limit the engine speed to a lower value until the filter clogging input becomes inactive. - either TCON will take no further action	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- check the wiring between the TCON and the filter indicator. - Check installation and operation of the filter indicator. - refer to the other reported errors to identify the affected analogue input
73.04	520326	4	Amber warning Flashing	Input Function: Filter Clogging: Voltage Below Normal, or Shorted To Low Source	TCON will disable inching . Other actions depends on parameter setting: - either TCON will limit the engine speed to a lower value until the filter clogging input becomes inactive. - either TCON will take no further action	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- check the wiring between the TCON and the filter indicator. - Check installation and operation of the filter indicator. - refer to the other reported errors to identify the affected analogue input
73.0F	520326	15	Amber warning Flashing	Input Function: Filter Clogging: Data Valid but Above Normal Operational Range - Least Severe	TCON will disable inching . Other actions depends on parameter setting: - either TCON will limit the engine speed to a lower value until the filter clogging input becomes inactive. - either TCON will take no further action	Heat the transmission	The measured value of the analog input is above the allowed range. This can be a normal condition due to cold operating temperature.	- heat the transmission and check if the fault code disappears.
74.00	520327	0	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Data Valid but Above Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The intermediate rpm is 100 rpm above what it should be. This can be determined using output and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.

74.01	520327	1	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Data Valid but Below Normal Operational Range - Most Severe	This message is given as additional information for transmission ratio faults. The action is taken by ratio fault monitoring .	Contact maintenance for troubleshooting	The intermediate rpm is 100 rpm below what it should be. This can be determined using output and turbine speed combined with the gear ratios. To be able to check this the turbine needs to be connected to the output.	Check the transmission, in order to understand if and which clutch is slipping . Check the transmission ratio settings of the controller.
74.05	520327	5	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Current Below Normal, or Open Circuit	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is below normal. The input most likely has a loose connection or it is shorted to battery plus (high source).	- Check the wiring between the TCON and the turbine speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input.
74.06	520327	6	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Current Above Normal or Grounded Circuit	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	The measured current in the speed sensor circuit is above normal. The input is most likely shorted to ground (low source).	- Check the wiring between the TCON and the intermediate speed sensor. - Check installation and operation of the speed sensor connected. - refer to the other reported errors to identify the affected speed input.
74.09	520327	9	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Abnormal Update Rate	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	No intermediate speed information available, as the message containing this information is missing on the CAN bus .	- Check if CAN message containing the intermediate speed value is sent correctly on the CAN bus . - Check if the repetition rate of that CAN message is fast enough.
74.13	520327	19	Amber Warning Solid	Speed Sensor Input Function: Intermediate 2 speed: Received Network Data in Error	The value of the intermediate speed sensor, will be replaced by an appropriate calculated value.	Contact maintenance for troubleshooting	The contents of the intermediate speed field in the CAN message is set to error or not available.	- Check CAN connection to the controller. - Check source of broadcast. - Check CAN connection to the controller.
12.00	520328	0	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
12.01	520328	1	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
12.03	520328	3	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
12.04	520328	4	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
12.09	520328	9	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Abnormal Update Rate	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
12.13	520328	19	Stop Lamp Flashing	Input Function: Clutch Low Pressure: Received Network Data in Error	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
13.00	520329	0	Stop Lamp Flashing	Input Function: Clutch High Pressure: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too high in comparison with the commanded pressure by TCON: - Proportional solenoid stuck. - Clutch failure	- Check clutch pressure at engine idle when clutch is NOT selected (+-0 bar required). - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
13.01	520329	1	Stop Lamp Flashing	Input Function: Clutch High Pressure: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	Measured pressure on this sensor is too low in comparison with the commanded pressure by TCON: - Supply pressure to transmission too low. - Proportional solenoid stuck. - Oil level too low. - Clutch failure	- Check oil level. - Check supply pressure to the transmission at engine idle. - Check clutch pressure at engine idle when clutch is selected. Log pressure (CAN trace or other measurement equipment) when problem occurs
13.03	520329	3	Stop Lamp Flashing	Input Function: Clutch High Pressure: Voltage Above Normal, or Shorted To High Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input most likely has a loose connection or is shorted to battery plus .	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.
13.04	520329	4	Stop Lamp Flashing	Input Function: Clutch High Pressure: Voltage Below Normal, or Shorted To Low Source	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is restarted.	Contact maintenance for troubleshooting	The input is most likely shorted to ground.	- Check the wiring between the TCON and the pressure sensor connected to this analog input. - Check installation and operation of the connected pressure sensor. - refer to the other reported errors to identify the configured analog input.

[illegible]

[illegible]

8A. 13	520347	19	Amber warning Flashing	Digital Input Function: Towing Feedback 1: Received Network Data in Error	TCON will continue normal operation, but the value of the input function Towing Feedback 1 will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function Towing Feedback 1 received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Towing Feedback 1 for requesting a valid value.
8B.09	520348	9	Amber warning Flashing	Digital Input Function: Towing Feedback 2: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
8B. 13	520348	19	Amber warning Flashing	Digital Input Function: Towing Feedback 2: Received Network Data in Error	TCON will continue normal operation, but the value of the input function Towing Feedback 2 will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function Towing Feedback 2 received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Towing Feedback 2 for requesting a valid value.
8C.02	520349	2	Stop Lamp Flashing	Digital Input Function: Redundant neutral: Data Erratic, Intermittent, or Incorrect	The value of the shifter function will be overruled by neutral selection. TCON will revert to shutdown mode.	Stop the vehicle and contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message or digital input contains an invalid or unavailable value.	Check the shift lever signals Check the redundant neutral signal.
8C.09	520349	9	Stop Lamp Flashing	Digital Input Function: Redundant neutral: Abnormal Update Rate	The value of the shifter function will be overruled by neutral selection. TCON will revert to shutdown mode.	Stop the vehicle and contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
8C. 13	520349	19	Stop Lamp Flashing	Digital Input Function: Redundant neutral: Received network data in Error	The value of the shifter function will be overruled by neutral selection. TCON will revert to shutdown mode.	Stop the vehicle and contact maintenance for troubleshooting	The expected value for the input function Redundant Neutral received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Redundant Neutral for requesting a valid value.
8D.09	520350	9	Amber warning Flashing	Digital Input Function: Eco Mode: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
8D. 13	520350	19	Amber warning Flashing	Digital Input Function: Eco Mode: Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Eco Mode" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Eco Mode" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Eco Mode for requesting a valid value.
8E.09	520351	9	Amber warning Flashing	Digital Input Function: Enhanced vehicle speed resolution: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
8E. 13	520351	19	Amber warning Flashing	Digital Input Function: Enhanced vehicle speed resolution: Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Enhanced vehicle speed resolution" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Enhanced vehicle speed resolution" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Enhanced vehicle speed resolution for requesting a valid value.
5A.0C	520352	12	Stop Lamp Flashing	Configuration Error: Incompatible Firmware: Failure Condition Exists	TCON will revert to TCON shutdown mode	Contact maintenance for troubleshooting .	During initialization the TCON detected that the firmware present is not compatible with this TCON. TCON is stuck in shutdown mode until this is solved	- download a valid and compatible firmware file (XML file) with the correct corresponding DANA tool. - if the problem persists, the program and/or dataflash in the TCON is defective: replace the TCON.
5A. 1F	520352	31	Stop Lamp Flashing	Configuration Error: Incompatible Data File: Failure Condition Exists	Refer to the other reported errors to get more information about the TCON action.	Perform controlled power down and restart the vehicle. If the problem persists, contact maintenance for troubleshooting .	During initialization the TCON detected that the flashsector 0 and/or 1 definitions-check does not correspond with what is expected. This might be due to a firmware upgrade.	A controlled power down of the TCON should be performed. If the error still persists, activate a working APT with the appropriate Dana tool, and perform a controlled power down again. If the error still persists, the TCON is probably damaged : replace the TCON. Refer to the other reported errors to get more information about the nature of the problem.
8F.09	520354	9	Amber warning Flashing	Digital Input Function: Loader Mode: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function "Loader Mode", received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
8F. 13	520354	19	Amber warning Flashing	Digital Input Function: Loader Mode Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Loader Mode" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Loader Mode" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Exhaust for requesting a valid value.
90.09	520355	9	Amber warning Flashing	Digital Input Function: Kickdown Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function "Kickdown", received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
90. 13	520355	19	Amber warning Flashing	Digital Input Function: Kickdown Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Kickdown" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Kickdown" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function "Kickdown" for requesting a valid value.
91.09	520356	9	Amber warning Flashing	Digital Input Function: Inching Mode Selection Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in reduced vehicle operation.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. Check the can bus communication.
91. 13	520356	19	Amber warning Flashing	Digital Input Function: Inching Mode Selection Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Inching Mode Select" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Inching mode select" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function "Inching mode select" for requesting a valid value.

92.09	520357	9	Amber warning Flashing	Digital Input Function: declutch enable/disable Selection Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in <u>reduced vehicle operation.</u>	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
92.13	520357	19	Amber warning Flashing	Digital Input Function: Declutch enable/disable Selection Received Network Data in Error	TCON will continue normal operation, but the value of the input function will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
93.09	520358	9	Amber warning Flashing	Digital Input Function: Inhibit Upshift: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in <u>reduced vehicle operation.</u>	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
93.13	520358	19	Amber warning Flashing	Digital Input Function: Inhibit Upshift: Received Network Data in Error	TCON will continue normal operation, but the value of the input function will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value.
82.09	520359	9	Amber warning Flashing	Digital Input Function: Neutral Lock: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in <u>reduced vehicle operation.</u>	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
82.13	520359	19	Amber warning Flashing	Digital Input Function: Neutral Lock: Received Network Data in Error	TCON will continue normal operation, but the value of the input function Operator Seated will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function Operator Seated received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Operator Seated for requesting a valid value.
83.09	520360	9	Amber warning Flashing	Digital Input Function: New Idle: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in <u>reduced vehicle operation.</u>	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
83.13	520360	19	Amber warning Flashing	Digital Input Function: New Idle: Received Network Data in Error	TCON will continue normal operation, but the value of the input function Operator Seated will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function Operator Seated received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Operator Seated for requesting a valid value.
60.09	520363	9	Amber warning Flashing	Digital Input Function: Upshift Command: Abnormal Update Rate	The TCON will flag the fault and the state of the function will be replaced by an appropriate default value. This may result in <u>reduced vehicle operation.</u>	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
60.13	520363	19	Amber warning Flashing	Digital Input Function: Upshift Command: Received Network Data in Error	TCON will continue normal operation, but the value of the input function "Eco Mode" will be replaced by the default value. This will result in reduced operation of the vehicle.	Contact maintenance for troubleshooting	The expected value for the input function "Upshift Command" received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Upshift Command for requesting a valid value.
61.09	520364	9	Stop Lamp Flashing	Digital Input Function: Motor status bit: Abnormal Update Rate	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	The expected value for the input function, received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function for requesting a valid value. <u>Check the can bus communication.</u>
61.13	520364	19	Stop Lamp Flashing	Digital Input Function: Motor status bit: Received Network Data in Error	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	The expected value for the input function received by the corresponding CAN message contains an invalid or unavailable value.	Check the source signal from the CAN message containing the input function Upshift Command for requesting a valid value.
1E.00	520368	0	Stop Lamp Flashing	Supply power for controller: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to recovery mode.	Contact maintenance for troubleshooting	Power supply to the controller too high: APC300 12V supply version: > 18 V APC300 24V supply version: > 32 V	- Check power supply - Check if a jump start setup is still connected.
1E.01	520368	1	Stop Lamp Flashing	Supply power for controller: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to recovery mode.	Contact maintenance for troubleshooting	Power supply to the controller too low (<10V)	- Check power supply to the controller. - Check alternator - Check the connection cables between the battery and the alternator - <u>Check the battery</u>
1E.0F	520368	15	Amber Warning Solid	Supply power for controller: Data Valid but Above Normal Operational Range - Least Severe	TCON will continue normal operation.	Contact maintenance for troubleshooting	Power supply to the controller too high, but not affecting normal operation	- Check power supply to the controller. - Check alternator - Check the connection cables between the battery and the alternator - <u>Check the battery</u>
1E.11	520368	17	Amber Warning Solid	Supply power for controller: Data Valid but Below Normal Operational Range - Least Severe	TCON will continue normal operation.	Contact maintenance for troubleshooting	Power supply to the controller too low, but not affecting normal operation	- Check power supply to the controller. - Check alternator - Check the connection cables between the battery and the alternator - <u>Check the battery</u>
1F.00	520369	0	Stop Lamp Solid	Supply power for sensors 8V: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	APC Internal voltage supply for the speed sensors : Vsense (8V) is below 7,2 V	- Check power supply. - check speed sensor connections - Send back controller for troubleshooting to <u>Dana Belgium N.V.</u>
1F.01	520369	1	Stop Lamp Solid	Supply power for sensors 8V: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	APC Internal voltage supply for the speed sensors : Vsense (8V) is above 8,8 V	- Check power supply. - check speed sensor connections - Send back controller for troubleshooting to <u>Dana Belgium N.V.</u>
77.00	520370	0	Stop Lamp Flashing	Supply power for sensors 5V: Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	APC Internal Sensor Reference exceeded the alarm limit. Correct measurement of TCON. L input signals can not be guaranteed anymore.	- Check power supply. - check connected sensors to the 5V supply - Send back controller for troubleshooting to <u>Dana Belgium N.V.</u>
77.01	520370	1	Stop Lamp Flashing	Supply power for sensors 5V: Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	APC Internal Sensor Reference exceeded the alarm limit. Correct measurement of TCON. L input signals can not be guaranteed anymore.	- Check power supply. - check connected sensors to the 5V supply - Send back controller for troubleshooting to <u>Dana Belgium N.V.</u>

76.05	520372	5	Stop Lamp Flashing	Ground sense fault: Current Below Normal, or Open Circuit	TCON will revert to transmission shutdown mode.	Contact maintenance for troubleshooting	APC. Sensor Reference ground faulty. Correct measurement of TCON input signals <u>can not be guaranteed anymore.</u>	Check ground pins, make sure ground sense is wired correctly.
22.1F	520379	31	Stop Lamp Flashing	redundant shutdown path sensors (RSP): Failure Condition Exists	TCON will revert to transmission limp home mode.	Contact maintenance for troubleshooting	One or more speed channel failures	- Check the wiring between the controller and the speed sensors. - <u>Check the speed sensors.</u>
22.01	520379	1	Stop Lamp Flashing	redundant shutdown path turbine (RSP): Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is <u>restarted.</u>	Stop the vehicle and contact maintenance for troubleshooting	Speed sensor failure, Supply pressure to transmission too low	- Check the wiring between the controller and the speed sensors. - Check the speed sensors. - Check the speed sensor positions . - <u>Check supply pressure to the transmission.</u>
22.00	520379	0	Stop Lamp Flashing	redundant shutdown path drum (RSP): Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is <u>restarted.</u>	Stop the vehicle and contact maintenance for troubleshooting	Speed sensor failure, Too much drag because of thick oil, Sticking or burnt clutch	- Check the wiring between the controller and the speed sensors. - Check the speed sensors. - Check the speed sensor positions .
22.00	520379	0	Stop Lamp Flashing	redundant shutdown path pressure (RSP): Data Valid but Above Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is <u>restarted.</u>	Stop the vehicle and contact maintenance for troubleshooting	Speed sensor failure, RSP drive solenoid failure. Supply pressure to transmission too low	- Check the wiring between the controller and the speed sensors. - Check the speed sensors. - Check the speed sensor positions . - Check supply pressure to the transmission, - <u>Check the RSP drive solenoid.</u>
22.01	520379	1	Stop Lamp Flashing	redundant shutdown path pressure (RSP): Data Valid but Below Normal Operational Range - Most Severe	TCON will revert to shutdown mode. This will result in inoperability of the vehicle. TCON will only resume normal operation once the error is cleared and TCON is <u>restarted.</u>	Stop the vehicle and contact maintenance for troubleshooting	Speed sensor failure, RSP drive solenoid failure. Too much drag because of thick oil, Sticking or burnt clutch	- Check the wiring between the controller and the speed sensors. - Check the speed sensors. - Check the speed sensor positions . - <u>Check the RSP drive solenoid.</u>
C1.09	520385	9	Amber warning Flashing	Can Message CVC_to_TC_1: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C2.09	520386	9	Amber warning Flashing	Can Message CVC_to_TC_2: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C3.09	520387	9	Amber warning Flashing	Can Message CVC_to_TC_6: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C4.09	520388	9	Amber warning Flashing	Can Message CVC_to_TC_4: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C5.09	520389	9	Amber warning Flashing	Can Message CVC_to_TC_5: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C6.09	520390	9	Amber warning Flashing	Can Message EEC1: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C7.09	520391	9	Amber warning Flashing	Can Message EEC2: Abnormal Update Rate	TCON will continue normal operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
C8.09	520392	9	Amber warning Flashing	Can Message TC1: Abnormal Update Rate	TCON will continue operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.

C9.09	520393	9	Amber warning Flashing	Can Message CCV51: Abnormal Update Rate	TCON will continue operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CA.09	520394	9	Amber warning Flashing	Can Message ERC1: Abnormal Update Rate	TCON will continue operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CB.09	520395	9	Amber warning Flashing	Can Message SSI: Abnormal Update Rate	TCON will continue operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CC.09	520396	9	Amber warning Flashing	Can Message EPD: Abnormal Update Rate	TCON will continue operation, but all the input functions contained in this CAN message will not be available. Refer to the other reported errors to see which functions are missing . These will determine the appropriate action taken by the TCON	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CD.09	520397	9	Amber warning Flashing	Can Message CM3: Abnormal Update Rate	TCON will continue operation, but the oil level measurement request will not be available.	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CE.09	520398	9	Stop Lamp Flashing	Can Message EM_to_TC_1: Abnormal Update Rate	TCON will revert to shutdown. As this message contains critical data, the vehicle cannot be driven until the message becomes available again.	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
CF.09	520399	9	Stop Lamp Flashing	Can Message AUXEM_to_TC_1: Abnormal Update Rate	TCON will revert to shutdown, as the auxiliary motor speed is a critical signal. The vehicle cannot be driven until the message becomes available again.	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
D0.00	520400	0	Amber warning Flashing	During calibration the turbine speed signal does not reach a minimum value.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting	At fill area calibration, timeout on full turbine pulldown.	
D0.02	520400	2	Amber warning Flashing	Transmission failure during calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	restart controller and check error reporting .	Transmission error detected	restart controller and check error reporting .
D0.07	520400	7	Amber warning Flashing	At start of P2mA, the pressure signal is too high.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting		
D0.08	520400	8	Amber warning Flashing	Measured pressure points do not allow interpolation.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting		
D0.0C	520400	12	Amber warning Flashing	During P2mA routine a pressure drop of more than 1 bar is detected.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting		
D0.10	520400	16	Amber warning Flashing	During calibration the turbine speed signal does not decrease.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting	Turbine speed doesn't decrease	
D0.11	520400	17	Amber warning Flashing	Pressure range for P2mA is too small. (< 2 bar)	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting		
D0.12	520400	18	Amber warning Flashing	during calibration the turbine speed signal is too low to start calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting	fill area calibration, at start of filling of clutch, the turbine speed is below parameter "Fill-TPD<SPD".	
D0.1F	520400	31	Amber warning Flashing	maximum number of retries reached during P2mA for the peak to peak pressure.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting		
D1.00	520401	0	Amber warning Flashing	During calibration the turbine speed signal does not reach a minimum value.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting	At fill area calibration, timeout on full turbine pulldown.	
D1.01	520401	1	Amber warning Flashing	Turbine speed signal does not reach a high enough value during release phase of the touch pressure calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	retry calibration, if still not successful contact maintenance for troubleshooting	turbine speed <= 5 rpm at end of release phase	
D1.02	520401	2	Amber warning Flashing	Transmission failure during calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed.	restart controller and check error reporting .	Transmission error detected	restart controller and check error reporting .

D1.07	520401	7	Amber warning Flashing	Turbine speed does not reach value = 0 rpm at end of closing phase of the touch pressure calibration	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	during touch pressure when at full pressure, the turbine speed > 5 rpm.	
D1.08	520401	11	Amber warning Flashing	No touch detection.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	timeout on detection of touch. (measured torque > (drag torque + parameter TouchTorq))	
D1.10	520401	16	Amber warning Flashing	During calibration the turbine speed signal does not decrease.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	Turbine speed doesn't decrease	
D1.12	520401	18	Amber warning Flashing	during calibration the turbine speed signal is too low to start calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	fill area calibration, at start of filling of clutch, the turbine speed is below parameter "Fill-TPD<SPD".	
D1.1F	520401	31	Amber warning Flashing	Early touch detection. Too much drag .	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	After filling phase and 2 seconds wait, the measured torque is bigger than parameter "DragTorqMax" or is bigger than parameter "TouchTorq". (if TouchTorq is bigger than 0. This is set to zero for TE30)	
D2.00	520402	0	Amber warning Flashing	During calibration the turbine speed signal does not reach a minimum value.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	At fill area calibration, timeout on full turbine pulldown.	
D2.02	520402	2	Amber warning Flashing	Transmission failure during calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	restart controller and check error reporting .	Transmission error detected	restart controller and check error reporting .
D2.07	520402	7	Amber warning Flashing	At start of fill area calibration, there's a timeout on closing the clutch in the other stage.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting		
D2.10	520402	16	Amber warning Flashing	During calibration the turbine speed signal does not decrease.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	Turbine speed doesn't decrease to start with turbine pulldown detection.	
D2.12	520402	18	Amber warning Flashing	during calibration the turbine speed signal is too low to start calibration.	TCON will hold calibration until the continue command is issued. Calibration can continue after this command, but this phase for this clutch will have failed	retry calibration, if still not successful contact maintenance for troubleshooting	fill area calibration, at start of filling of clutch, the turbine speed is below parameter "Fill-TPD<SPD".	
DA.09	520410	9	Amber warning Flashing	Can Message Em_ Ready: Abnormal Update Rate	TCON will revert to restricted mode until the message becomes available. The electro motor ready state is defaulted to "not ready" and this means error reporting is put on hold and no driving is possible	Contact maintenance for troubleshooting	The CAN message is not received within the expected timeframe.	- Check if the CAN message is sent correctly on the CAN bus . - Check if the repetition rate of the CAN message is fast enough.
E5.0D	520416	13	Amber Warning Solid	Exceed Transmission Calibration Interval: Out of Calibration	TCON will continue normal operation, but the calibration values of the transmission will be replaced by the default values. This may result in reduced performance quality of the vehicle	Perform a transmission calibration or Contact maintenance for troubleshooting .	The transmission calibration in the TCON has not been performed yet. The corresponding exceed code is reported to indicate this situation to the driver.	Perform a transmission calibration.
E5.0F	520416	15	Amber Warning Solid	Exceed Transmission Calibration Interval: Data Valid but Above Normal Operational Range - Least Severe	TCON will continue normal operation, but the calibration values of the transmission are not up to date. This may result in reduced performance quality of the vehicle	Perform a transmission calibration.	The transmission calibration in the TCON has not been performed yet or has expired. The corresponding exceed code is reported to indicate this situation to the driver.	Perform a transmission calibration.
E0.0F	520417	15	Protection Lamp	Exceed Downshift Vehicle Speed: Data Valid but Above Normal Operational Range - Least Severe	No special action taken by the TCON. Normal application logics are active to handle the situation.	Slow down the vehicle to decrease the speed until the requested downshift is allowed.	An operational limit was exceeded during normal operation. The corresponding exceed code is reported to indicate this situation to the driver	none
E1.0F	520418	15	Protection Lamp	Exceed Direction Change Vehicle Speed: Data Valid but Above Normal Operational Range - Least Severe	No special action taken by the TCON. Normal application logics are active to handle the situation.	Slow down the vehicle to decrease the speed until the requested direction change is allowed.	An operational limit was exceeded during normal operation. The corresponding exceed code is reported to indicate this situation to the driver	none
E2.0F	520419	15	Protection Lamp	Exceed Direction Change Engine Speed: Data Valid but Above Normal Operational Range - Least Severe	No special action taken by the TCON. Normal application logics are active to handle the situation.	Lower the engine speed until the requested direction change is allowed.	An operational limit was exceeded during normal operation. The corresponding exceed code is reported to indicate this situation to the driver	none
E3.0F	520420	15	Protection Lamp	Exceed Reduced vehicle speed : Data Valid but Above Normal Operational Range - Least Severe	TCON is limiting the vehicle speed to the set vehicle speed limit. Speed is within 85%-105% range of set speed limit. (the 85% is a configurable, and may deviate on some applications)	No action is requested, the vehicle speed is limited due to request.	The vehicle speed limitation is active.	none
E3.10	520420	16	Protection Lamp	Exceed Reduced vehicle speed : Data Valid but Above Normal Operational Range - Moderately Severe	TCON is limiting the vehicle speed to the maximum or reduced vehicle speed limit, but detects a higher vehicle speed	Slow down the vehicle, until that the vehicle speed comes below the maximum or reduced vehicle speed limit	The vehicle speed is above the maximum vehicle speed limit.	none
E4.0F	520421	15	Protection Lamp	Exceed Abnormal Acceleration/Deceleration: Data Valid but Above Normal Operational Range - Least Severe	An abnormal acceleration is detected.	The driver should slow down the engine, in order that the wheels can not spin any more.	The controller detected an abnormal acceleration, and will disable the automatic shifting for a certain time .	none
E4.11	520421	17	Protection Lamp	Exceed Abnormal Acceleration/Deceleration: Data Valid but Below Normal Operational Range - Least Severe	An abnormal deceleration is detected.	The driver should slow down, in order that the wheels can not slip any more.	The controller detected an abnormal deceleration, and will disable the automatic shifting for a certain time .	none
E6.0E	520422	14	Protection Lamp	Exceed Operator seated conditions: Special Instruction (consult documentation)	Operator is not seated with shiftlever not in neutral and/or the parking brake not activated.	The driver should take place in the operator seat and confirm his presence (cycle the shiftlever through neutral and/or release the throttle pedal)	The controller detected that the operator is not seated with the shiftlever not in neutral and/or the parking brake not activated.	- Check operator seat switch - Check parking brake - Check shift lever
E7.0F	520423	15	Protection Lamp	Exceed transmission speed limits: Data Valid but Above Normal Operational Range - Least Severe	TCON will force engine idle to prevent damage.	Slow down the vehicle, in order that the speed drops below the maximum speed limit.	TCON detects that the turbine, intermediate or output speed is above its limit	none

E8.0F	520424	15	Protection Lamp	Exceed Low High range conditions: Data Valid but Above Normal Operational Range - Least Severe	Condition to change low high range selector not fulfilled. Vehicle has to stand still.	Brake and stop the vehicle until this exceed code is no longer present.	TCON detected that the vehicle is still moving.	none
E9.0F	520425	15	Protection Lamp	Exceed 4WD/2WD drive conditions: Data Valid but Above Normal Operational Range - Least Severe	Condition to change from 4wd to 2wd or visa versa not fulfilled. Vehicle has to stand still.	Brake and stop the vehicle until this exceed code is no longer present.	TCON detected that the vehicle is still moving.	none
EA.0F	520426	15	Protection Lamp	Exceed Cutch Energy: Data Valid but Above Normal Operational Range - Least Severe	Depending on the clutch state, the TCON. H312 has closed or opened the clutch without further modulation.	No action needed	Clutch closing/slipping/inching took too long.	none
EB. 11	520427	17	Protection Lamp	Exceed code temperature not in operating range; Data Valid but Below Normal Operational Range - Least Severe	Operating temperature not reached. Engine throttle is limited.	Heat up transmission.	Operating temperature not reached	none
F0. 12	520432	18	Stop Lamp Flashing	Controller boots in restricted mode.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	The controller encountered an unexpected error. Most likely this is due to a bad ground sense.	- If this occurs right after performing a firmware update of the TCON. H, this should be solved by performing a normal power down of the TCON. H. - Check the Ground Sense line (pin 60)
F3.00	520435	0	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 0.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 0 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.01	520435	1	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 1.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 1 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.02	520435	2	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 2.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 2 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.03	520435	3	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 3.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 3 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.04	520435	4	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 4.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 4 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.05	520435	5	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 5.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 5 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.06	520435	6	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 6.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 6 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.07	520435	7	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 7.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 7 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.08	520435	8	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 8.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting.	During initialization the TCON detected that flashgroup 8 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)

F3.09	520435	9	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 9.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting .	During initialization the TCON detected that flashgroup 9 is incomplete or partially corrupt. TCON is stuck in shutdown mode until this is solved.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F3.0A	520435	10	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 10.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting .	During initialization the TCON detected that the logging part of the data flash is corrupt. All functional logging data (like travelled distance etc..) but also error logging data will be lost and reset to zero.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, contact Dana Belgium N.V.
F3.0B	520435	11	Stop Lamp Flashing	APC Data Flash partially corrupt: parameter group 11.	TCON reverts to shutdown mode.	Turn power off and restart the vehicle. If the problem persists, contact maintenance for troubleshooting .	During initialization the TCON detected that the logging part of the data flash is corrupt. All functional logging data (like travelled distance etc..) but also error logging data will be lost and reset to zero.	- If this occurs right after performing a firmware update of the TCON, this should be solved by performing a normal power down of the TCON. - If the problem persists, contact Dana Belgium N.V.
F4.00	520436	0	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 0.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 0 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.01	520436	1	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 1.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 1 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.02	520436	2	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 2.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 2 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.03	520436	3	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 3.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 3 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.04	520436	4	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 4.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 4 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.05	520436	5	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 5.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 5 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.06	520436	6	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 6.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 6 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.07	520436	7	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 7.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 7 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.08	520436	8	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 8.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 8 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F4.09	520436	9	Stop Lamp Flashing	APC Data Flash permanently corrupt: parameter group 9.	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	During initialization the TCON detected that a critical part of the data flash group 9 is corrupt. TCON is stuck in shutdown mode.	- Download an APT file compatible with the active firmware into the TCON. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F5.00	520437	0	Stop Lamp Flashing	Dana Configuration Error	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	This could mean that an invalid configuration set is selected.	- Check the activated configuration set for this vehicle.
F5. 12	520437	18	Stop Lamp Flashing	ECU state unsafe	TCON reverts to shutdown mode.	Contact maintenance for troubleshooting .	This can be the result of a faulty wiring . The power supply for the ECU is not correct.	- Check the voltage connections on the TCON. Most likely pin 60 (ground) is not connected or faulty. - If the error code still persists, the data flash may be corrupt (contact Dana Belgium N.V.)
F7.00	520439	0	Amber warning Flashing	Severe engine limitation active due to low temperature.	TCON limits the engine to the lowest limit.	Heat up transmission.		
F7.01	520439	1	Amber warning Flashing	Engine limitation active due to low temperature.	TCON limits the engine.	Heat up transmission.		
F8.00	520440	0	Amber warning Flashing	Direction selection not available due to low temperature.	TCON prevents direction engagement until the converter out and sump temperature is in the operating range.	Heat up transmission.		
F8.01	520440	1	Stop Lamp Flashing	No transmission oil pressure.	TCON engine shutdown request.	Shut down the engine, heat up transmission by heater.	insufficient oil pressure was detected.	